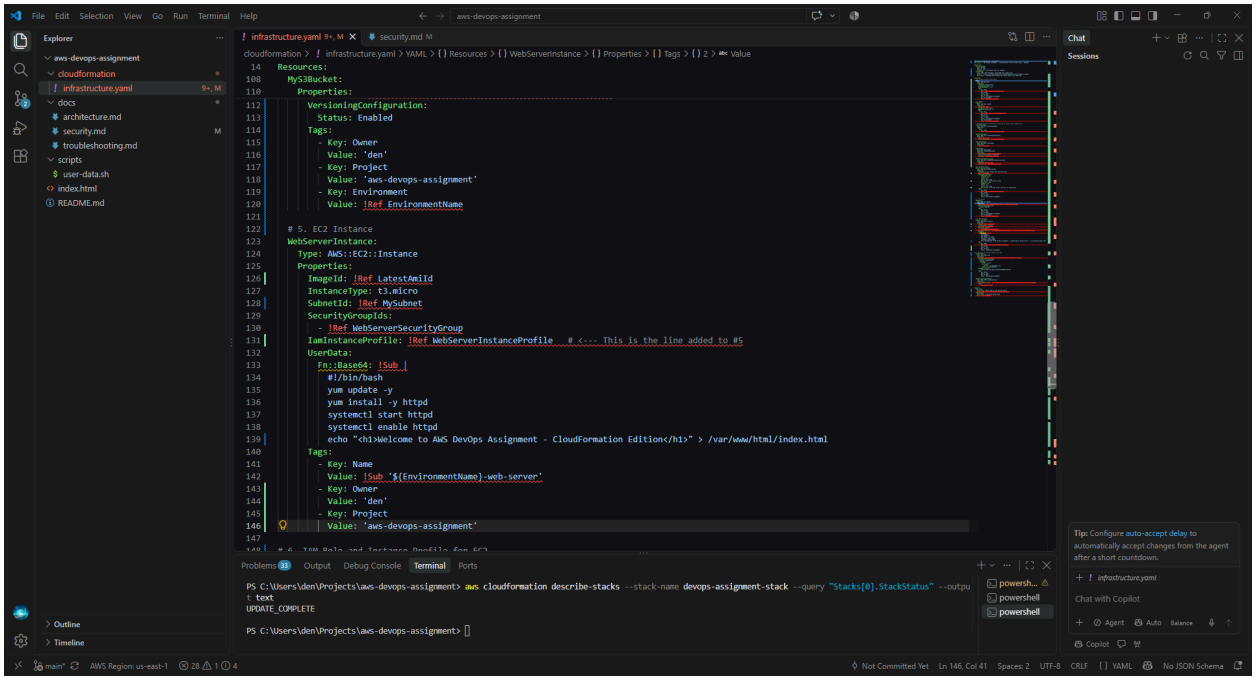
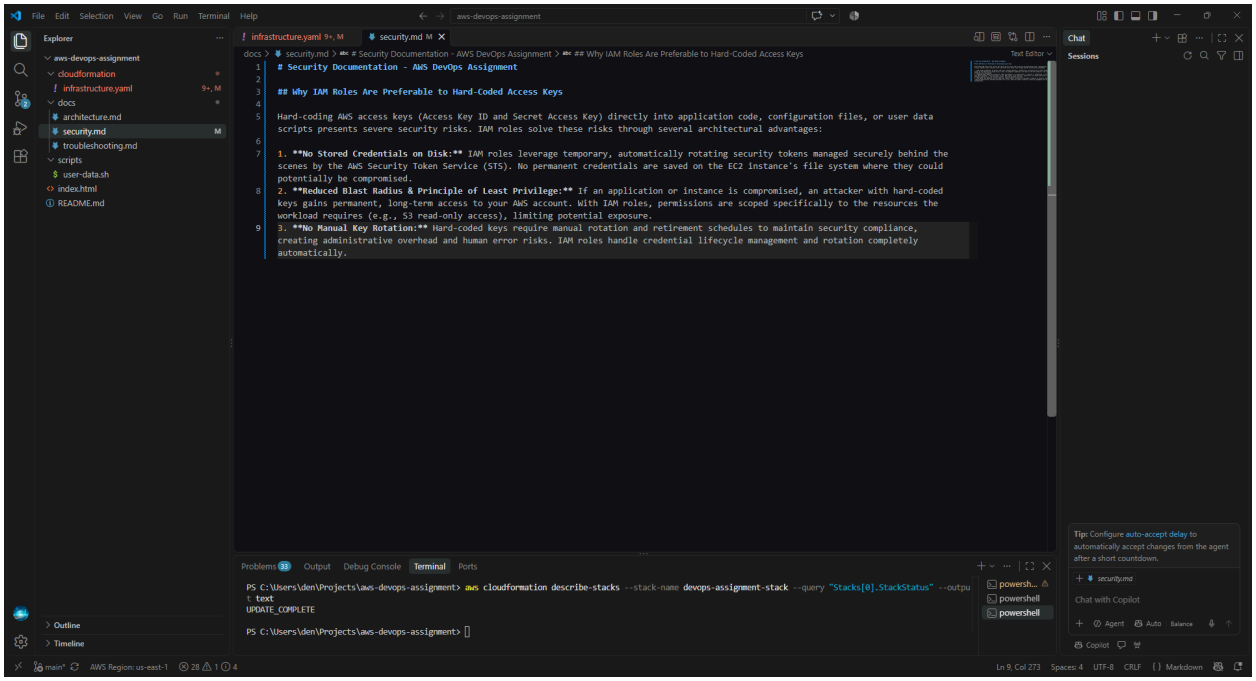
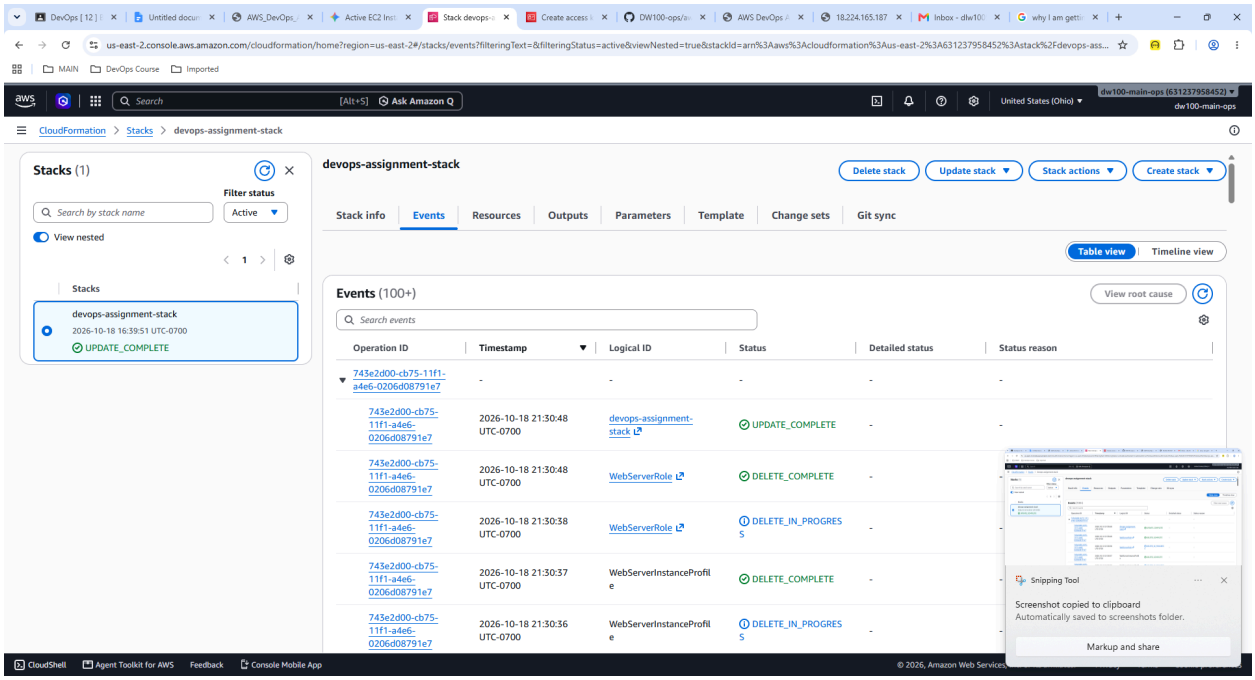

File Edit Selection View Go Run Terminal Help
aws-devops-assignment
Explorer
aws-devops-assignment
cloudformation
infrastructure.yaml 9+, M
docs
architecture.md
security.md M
troubleshooting.md
scripts
user-data.sh
index.html
README.md
infrastructure.yaml 9+, M
security.md M
cloudformation > infrastructure.yaml > YAML > { } Resources > { } WebServerInstance > { } Properties > [ ] Tags > { } 2 > abc Value
14 Resources:
108 MyS3Bucket:
110 Properties:
112 VersioningConfiguration:
113 Status: Enabled
114 Tags:
115 - Key: Owner
116 Value: 'den'
117 - Key: Project
118 Value: 'aws-devops-assignment'
119 - Key: Environment
120 Value: !Ref EnvironmentName
122 # 5. EC2 Instance
123 WebServerInstance:
124 Type: AWS::EC2::Instance
125 Properties:
126 ImageId: !Ref LatestAmiId
127 InstanceType: t3.micro
128 SubnetId: !Ref MySubnet
129 SecurityGroupIds:
130 - !Ref WebServerSecurityGroup
131 IamInstanceProfile: !Ref WebServerInstanceProfile # <---- This is the line added to #5
132 UserData:
133 Fn::Base64: !Sub |
134 #!/bin/bash
135 yum update -y
136 yum install -y httpd
137 systemctl start httpd
138 systemctl enable httpd
139 echo "<h1>Welcome to AWS DevOps Assignment - CloudFormation Edition</h1>" > /var/www/html/index.html
140 Tags:
141 - Key: Name
142 Value: !Sub '${EnvironmentName}-web-server'
143 - Key: Owner
144 Value: 'den'
145 - Key: Project
146 Value: 'aws-devops-assignment'
147
Problems 33 Output Debug Console Terminal Ports
PS C:\Users\den\Projects\aws-devops-assignment> aws cloudformation describe-stacks --stack-name devops-assignment-stack --query "Stacks[0].StackStatus" --output text
UPDATE_COMPLETE
PS C:\Users\den\Projects\aws-devops-assignment>
powersh...
powershell
powershell
Outline
Timeline
Chat
Sessions
Tip: Configure auto-accept delay to automatically accept changes from the agent after a short countdown.
infrastructure.yaml
Chat with Copilot
Agent Auto Balance
Copilot
main* AWS Region: us-east-1 28 1 4
Not Committed Yet Ln 146, Col 41 Spaces: 2 UTF-8 CRLF YAML No JSON Schema


File Edit Selection View Go Run Terminal Help
aws-devops-assignment
Explorer
aws-devops-assignment
cloudformation
infrastructure.yaml 9+, M
docs
architecture.md
security.md M
troubleshooting.md
scripts
user-data.sh
index.html
README.md
infrastructure.yaml 9+, M
security.md M
docs > security.md > # Security Documentation - AWS DevOps Assignment > ## Why IAM Roles Are Preferable to Hard-Coded Access Keys
Text Editor
1 # Security Documentation - AWS DevOps Assignment
2
3 ## Why IAM Roles Are Preferable to Hard-Coded Access Keys
4
5 Hard-coding AWS access keys (Access Key ID and Secret Access Key) directly into application code, configuration files, or user data scripts presents severe security risks. IAM roles solve these risks through several architectural advantages:
6
7 1. **No Stored Credentials on Disk:** IAM roles leverage temporary, automatically rotating security tokens managed securely behind the scenes by the AWS Security Token Service (STS). No permanent credentials are saved on the EC2 instance's file system where they could potentially be compromised.
8 2. **Reduced Blast Radius & Principle of Least Privilege:** If an application or instance is compromised, an attacker with hard-coded keys gains permanent, long-term access to your AWS account. With IAM roles, permissions are scoped specifically to the resources the workload requires (e.g., S3 read-only access), limiting potential exposure.
9 3. **No Manual Key Rotation:** Hard-coded keys require manual rotation and retirement schedules to maintain security compliance, creating administrative overhead and human error risks. IAM roles handle credential lifecycle management and rotation completely automatically.
Problems 33 Output Debug Console Terminal Ports
PS C:\Users\den\Projects\aws-devops-assignment> aws cloudformation describe-stacks --stack-name devops-assignment-stack --query "Stacks[0].StackStatus" --output text
UPDATE_COMPLETE
PS C:\Users\den\Projects\aws-devops-assignment>
powersh...
powershell
powershell
Outline
Timeline
Chat
Sessions
Tip: Configure auto-accept delay to automatically accept changes from the agent after a short countdown.
security.md
Chat with Copilot
Agent Auto Balance
Copilot
main* AWS Region: us-east-1 28 1 4
Ln 9, Col 273 Spaces: 4 UTF-8 CRLF Markdown


us-east-2.console.aws.amazon.com/cloudformation/home?region=us-east-2#/stacks/events?filteringText=&filteringStatus=active&viewNested=true&stackId=arn%3Aaws%3Acloudformation%3Aus-east-2%3A631237958452%3Astack%2Fdevops-ass...
CloudFormation > Stacks > devops-assignment-stack
Stacks (1)
Filter status
Search by stack name
Active
View nested
Stacks
devops-assignment-stack
2026-10-18 16:39:51 UTC-0700
UPDATE_COMPLETE
devops-assignment-stack
Delete stack
Update stack
Stack actions
Create stack
Stack info
Events
Resources
Outputs
Parameters
Template
Change sets
Git sync
Table view
Timeline view
Events (100+)
View root cause
Search events
Operation ID
Timestamp
Logical ID
Status
Detailed status
Status reason
743e2d00-cb75-11f1-a4e6-0206d08791e7
2026-10-18 21:30:48 UTC-0700
devops-assignment-stack
UPDATE_COMPLETE
WebServerRole
DELETE_COMPLETE
DELETE_IN_PROGRESS
WebServerInstanceProfile
2026-10-18 21:30:38 UTC-0700
2026-10-18 21:30:37 UTC-0700
2026-10-18 21:30:36 UTC-0700
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App